

and $SSR_1 = \sum_{t=m+1}^T \hat{e}_t^2$, where $\hat{e}_t$ is the least squares residual of the prior linear regression. Then we have

$$F = \frac{(SSR_0 - SSR_1)/m}{SSR_1/(T - 2m - 1)},$$

which is asymptotically distributed as a chi-squared distribution with m degrees of freedom under the null hypothesis. The decision rule is to reject the null hypothesis if $F > \chi_m^2(\alpha)$, where $\chi_m^2(\alpha)$ is the upper $100(1 - \alpha)$ th percentile of χ_m^2 , or the p -value of F is less than α .

To demonstrate, we consider the monthly log stock returns of Intel Corporation from 1973 to 2003; see Example 3.1 below. The series does not have significant serial correlations so that it can be directly used to test for the ARCH effect. Indeed, the $Q(m)$ statistics of the return series give $Q(12) = 18.57$ with p -value 0.10, confirming no serial correlations in the data. On the other hand, the Lagrange multiplier test shows strong ARCH effects with test statistic $F \approx 43.5$, the p -value of which is close to zero.

S-Plus Demonstration

Denote the return series by `intc`. Note that the command `archTest` applies directly to the a_t series, not to a_t^2 .

```
> autocorTest(intc, lag=12)
Test for Autocorrelation: Ljung-Box
Null Hypothesis: no autocorrelation

Test Statistics:
Test Stat 18.5664  p.value  0.0995

Dist. under Null: chi-square with 12 degrees of freedom
Total Observ.: 372

> archTest(intc, lag=12)
Test for ARCH Effects: LM Test
Null Hypothesis: no ARCH effects

Test Statistics:
Test Stat 43.5041  p.value  0.0000

Dist. under Null: chi-square with 12 degrees of freedom
```

3.4 THE ARCH MODEL

The first model that provides a systematic framework for volatility modeling is the ARCH model of Engle (1982). The basic idea of ARCH models is that (a) the shock a_t of an asset return is serially uncorrelated, but dependent, and (b) the

dependence of a_t can be described by a simple quadratic function of its lagged values. Specifically, an ARCH(m) model assumes that

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2, \quad (3.5)$$

where $\{\epsilon_t\}$ is a sequence of independent and identically distributed (iid) random variables with mean zero and variance 1, $\alpha_0 > 0$, and $\alpha_i \geq 0$ for $i > 0$. The coefficients α_i must satisfy some regularity conditions to ensure that the unconditional variance of a_t is finite. In practice, ϵ_t is often assumed to follow the standard normal or a standardized Student- t distribution or a generalized error distribution.

From the structure of the model, it is seen that large past squared shocks $\{a_{t-i}^2\}_{i=1}^m$ imply a large conditional variance σ_t^2 for the innovation a_t . Consequently, a_t tends to assume a large value (in modulus). This means that, under the ARCH framework, large shocks tend to be followed by another large shock. Here I use the word *tend* because a large variance does not necessarily produce a large realization. It only says that the probability of obtaining a large variate is greater than that of a smaller variance. This feature is similar to the volatility clusterings observed in asset returns.

The ARCH effect also occurs in other financial time series. Figure 3.2 shows the time plots of (a) the percentage changes in Deutsche mark/U.S. dollar exchange rate measured in 10-minute intervals from June 5, 1989 to June 19, 1989 for 2488

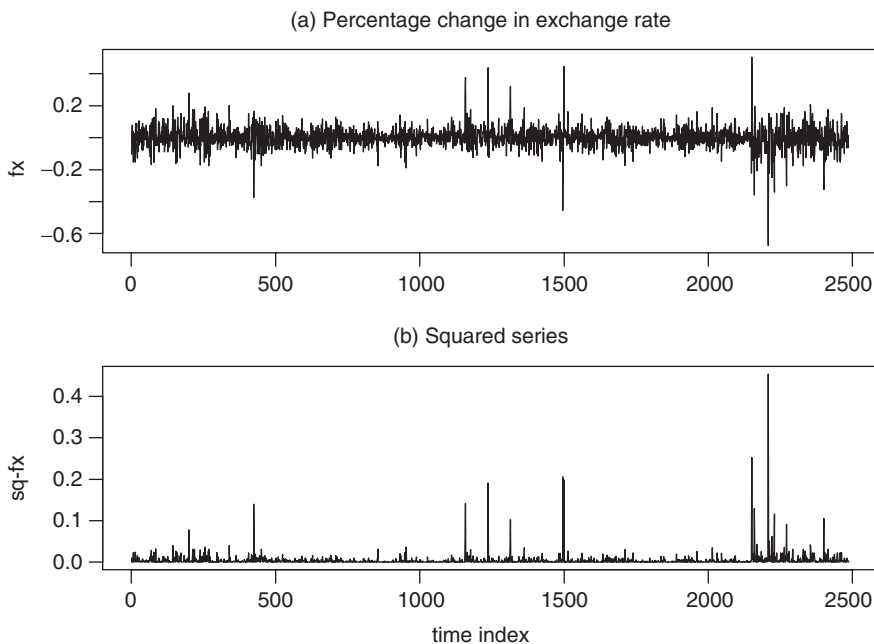


Figure 3.2. (a) Time plot of 10-minute returns of the exchange rate between Deutsche mark and U.S. dollar and (b) the squared returns.

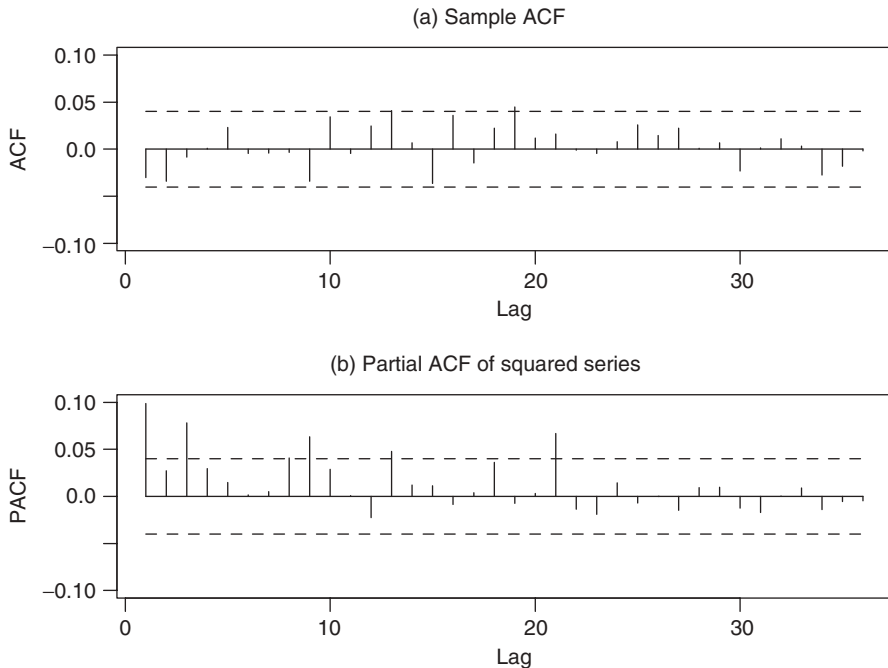


Figure 3.3. (a) Sample autocorrelation function of the return series of mark/dollar exchange rate and (b) sample partial autocorrelation function of the squared returns.

observations, and (b) the squared series of the percentage changes. Big percentage changes occurred occasionally, but there were certain stable periods. Figure 3.3a shows the sample ACF of the percentage change series. Clearly, the series has no serial correlation. Figure 3.3b shows the sample PACF of the squared series of percentage change. It is seen that there are some big spikes in the PACF. Such spikes suggest that the percentage changes are not serially independent and have some ARCH effects.

Remark. Some authors use h_t to denote the conditional variance in Eq. (3.5). In this case, the shock becomes $a_t = \sqrt{h_t}\epsilon_t$. $\square$

3.4.1 Properties of ARCH Models

To understand the ARCH models, it pays to carefully study the ARCH(1) model

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2,$$

where $\alpha_0 > 0$ and $\alpha_1 \geq 0$. First, the unconditional mean of a_t remains zero because

$$E(a_t) = E[E(a_t|F_{t-1})] = E[\sigma_t E(\epsilon_t)] = 0.$$

Second, the unconditional variance of a_t can be obtained as

$$\begin{aligned}\text{Var}(a_t) &= E(a_t^2) = E[E(a_t^2|F_{t-1})] \\ &= E(\alpha_0 + \alpha_1 a_{t-1}^2) = \alpha_0 + \alpha_1 E(a_{t-1}^2).\end{aligned}$$

Because a_t is a stationary process with $E(a_t) = 0$, $\text{Var}(a_t) = \text{Var}(a_{t-1}) = E(a_{t-1}^2)$. Therefore, we have $\text{Var}(a_t) = \alpha_0 + \alpha_1 \text{Var}(a_t)$ and $\text{Var}(a_t) = \alpha_0/(1 - \alpha_1)$. Since the variance of a_t must be positive, we require $0 \leq \alpha_1 < 1$. Third, in some applications, we need higher order moments of a_t to exist and, hence, α_1 must also satisfy some additional constraints. For instance, to study its tail behavior, we require that the fourth moment of a_t is finite. Under the normality assumption of ϵ_t in Eq. (3.5), we have

$$E(a_t^4|F_{t-1}) = 3[E(a_t^2|F_{t-1})]^2 = 3(\alpha_0 + \alpha_1 a_{t-1}^2)^2.$$

Therefore,

$$E(a_t^4) = E[E(a_t^4|F_{t-1})] = 3E(\alpha_0 + \alpha_1 a_{t-1}^2)^2 = 3E(\alpha_0^2 + 2\alpha_0\alpha_1 a_{t-1}^2 + \alpha_1^2 a_{t-1}^4).$$

If a_t is fourth-order stationary with $m_4 = E(a_t^4)$, then we have

$$\begin{aligned}m_4 &= 3[\alpha_0^2 + 2\alpha_0\alpha_1 \text{Var}(a_t) + \alpha_1^2 m_4] \\ &= 3\alpha_0^2 \left(1 + 2\frac{\alpha_1}{1 - \alpha_1}\right) + 3\alpha_1^2 m_4.\end{aligned}$$

Consequently,

$$m_4 = \frac{3\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}.$$

This result has two important implications: (a) since the fourth moment of a_t is positive, we see that α_1 must also satisfy the condition $1 - 3\alpha_1^2 > 0$; that is, $0 \leq \alpha_1^2 < \frac{1}{3}$; and (b) the unconditional kurtosis of a_t is

$$\frac{E(a_t^4)}{[\text{Var}(a_t)]^2} = 3 \frac{\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)} \times \frac{(1 - \alpha_1)^2}{\alpha_0^2} = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} > 3.$$

Thus, the excess kurtosis of a_t is positive and the tail distribution of a_t is heavier than that of a normal distribution. In other words, the shock a_t of a conditional Gaussian ARCH(1) model is more likely than a Gaussian white noise series to produce “outliers.” This is in agreement with the empirical finding that “outliers” appear more often in asset returns than that implied by an iid sequence of normal random variates.

These properties continue to hold for general ARCH models, but the formulas become more complicated for higher order ARCH models. The condition $\alpha_i \geq 0$ in Eq. (3.5) can be relaxed. It is a condition to ensure that the conditional variance σ_t^2

is positive for all t . In fact, a natural way to achieve positiveness of the conditional variance is to rewrite an ARCH(m) model as

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + A'_{m,t-1} \Omega A_{m,t-1}, \quad (3.6)$$

where $A_{m,t-1} = (a_{t-1}, \dots, a_{t-m})'$ and Ω is an $m \times m$ non-negative definite matrix. The ARCH(m) model in Eq. (3.5) requires Ω to be diagonal. Thus, Engle's model uses a parsimonious approach to approximate a quadratic function. A simple way to achieve Eq. (3.6) is to employ a random-coefficient model for a_t ; see the CHARMA and RCA models discussed later.

3.4.2 Weaknesses of ARCH Models

The advantages of ARCH models include properties discussed in the previous subsection. The model also has some weaknesses:

1. The model assumes that positive and negative shocks have the same effects on volatility because it depends on the square of the previous shocks. In practice, it is well known that price of a financial asset responds differently to positive and negative shocks.
2. The ARCH model is rather restrictive. For instance, α_1^2 of an ARCH(1) model must be in the interval $[0, \frac{1}{3}]$ if the series has a finite fourth moment. The constraint becomes complicated for higher order ARCH models. In practice, it limits the ability of ARCH models with Gaussian innovations to capture excess kurtosis.
3. The ARCH model does not provide any new insight for understanding the source of variations of a financial time series. It merely provides a mechanical way to describe the behavior of the conditional variance. It gives no indication about what causes such behavior to occur.
4. ARCH models are likely to overpredict the volatility because they respond slowly to large isolated shocks to the return series.

3.4.3 Building an ARCH Model

Among volatility models, specifying an ARCH model is relatively easy. Details are given below.

Order Determination

If an ARCH effect is found to be significant, one can use the PACF of a_t^2 to determine the ARCH order. Using PACF of a_t^2 to select the ARCH order can be justified as follows. From the model in Eq. (3.5), we have

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \dots + \alpha_m a_{t-m}^2.$$

For a given sample, a_t^2 is an unbiased estimate of σ_t^2 . Therefore, we expect that a_t^2 is linearly related to $a_{t-1}^2, \dots, a_{t-m}^2$ in a manner similar to that of an autoregressive

model of order m . Note that a single a_t^2 is generally not an efficient estimate of σ_t^2 , but it can serve as an approximation that could be informative in specifying the order m .

Alternatively, define $\eta_t = a_t^2 - \sigma_t^2$. It can be shown that $\{\eta_t\}$ is an un-correlated series with mean 0. The ARCH model then becomes

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2 + \eta_t,$$

which is in the form of an AR(m) model for a_t^2 , except that $\{\eta_t\}$ is not an iid series. From Chapter 2, PACF of a_t^2 is a useful tool to determine the order m . Because $\{\eta_t\}$ are not identically distributed, the least squares estimates of the prior model are consistent, but not efficient. The PACF of a_t^2 may not be effective when the sample size is small.

Estimation

Three likelihood functions are commonly used in ARCH estimation. Under the normality assumption, the likelihood function of an ARCH(m) model is

$$\begin{aligned} f(a_1, \dots, a_T | \boldsymbol{\alpha}) &= f(a_T | F_{T-1}) f(a_{T-1} | F_{T-2}) \cdots f(a_{m+1} | F_m) f(a_1, \dots, a_m | \boldsymbol{\alpha}) \\ &= \prod_{t=m+1}^T \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{a_t^2}{2\sigma_t^2}\right) \times f(a_1, \dots, a_m | \boldsymbol{\alpha}), \end{aligned}$$

where $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \dots, \alpha_m)'$ and $f(a_1, \dots, a_m | \boldsymbol{\alpha})$ is the joint probability density function of $a_1, \dots, a_m$. Since the exact form of $f(a_1, \dots, a_m | \boldsymbol{\alpha})$ is complicated, it is commonly dropped from the prior likelihood function, especially when the sample size is sufficiently large. This results in using the conditional likelihood function

$$f(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = \prod_{t=m+1}^T \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{a_t^2}{2\sigma_t^2}\right),$$

where σ_t^2 can be evaluated recursively. We refer to estimates obtained by maximizing the prior likelihood function as the conditional maximum likelihood estimates (MLEs) under normality.

Maximizing the conditional likelihood function is equivalent to maximizing its logarithm, which is easier to handle. The conditional log likelihood function is

$$\ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = \sum_{t=m+1}^T \left(-\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma_t^2) - \frac{1}{2} \frac{a_t^2}{\sigma_t^2} \right).$$

Since the first term $\ln(2\pi)$ does not involve any parameters, the log likelihood function becomes

$$\ell(a_{m+1}, \dots, a_T | \boldsymbol{\alpha}, a_1, \dots, a_m) = - \sum_{t=m+1}^T \left(\frac{1}{2} \ln(\sigma_t^2) + \frac{1}{2} \frac{a_t^2}{\sigma_t^2} \right),$$

where $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2$ can be evaluated recursively.

In some applications, it is more appropriate to assume that ϵ_t follows a heavy-tailed distribution such as a standardized Student- t distribution. Let x_v be a Student- t distribution with v degrees of freedom. Then $\text{Var}(x_v) = v/(v-2)$ for $v > 2$, and we use $\epsilon_t = x_v/\sqrt{v/(v-2)}$. The probability density function of ϵ_t is

$$f(\epsilon_t|v) = \frac{\Gamma((v+1)/2)}{\Gamma(v/2)\sqrt{(v-2)\pi}} \left(1 + \frac{\epsilon_t^2}{v-2}\right)^{-(v+1)/2}, \quad v > 2, \quad (3.7)$$

where $\Gamma(x)$ is the usual gamma function (i.e., $\Gamma(x) = \int_0^\infty y^{x-1} e^{-y} dy$). Using $a_t = \sigma_t \epsilon_t$, we obtain the conditional likelihood function of a_t as

$$f(a_{m+1}, \dots, a_T | \alpha, A_m) = \prod_{t=m+1}^T \frac{\Gamma((v+1)/2)}{\Gamma(v/2)\sqrt{(v-2)\pi}} \frac{1}{\sigma_t} \left(1 + \frac{a_t^2}{(v-2)\sigma_t^2}\right)^{-(v+1)/2},$$

where $v > 2$ and $A_m = (a_1, a_2, \dots, a_m)$. We refer to the estimates that maximize the prior likelihood function as the conditional MLEs under t -distribution. The degrees of freedom of the t -distribution can be specified a priori or estimated jointly with other parameters. A value between 3 and 6 is often used if it is prespecified.

If the degrees of freedom v of the Student- t distribution is prespecified, then the conditional log likelihood function is

$$\ell(a_{m+1}, \dots, a_T | \alpha, A_m) = - \sum_{t=m+1}^T \left[\frac{v+1}{2} \ln \left(1 + \frac{a_t^2}{(v-2)\sigma_t^2} \right) + \frac{1}{2} \ln(\sigma_t^2) \right]. \quad (3.8)$$

If one wishes to estimate v jointly with other parameters, then the log likelihood function becomes

$$\begin{aligned} \ell(a_{m+1}, \dots, a_T | \alpha, v, A_m) \\ = (T-m)[\ln(\Gamma((v+1)/2)) - \ln(\Gamma(v/2)) - 0.5 \ln((v-2)\pi)] \\ + \ell(a_{m+1}, \dots, a_T | \alpha, A_m), \end{aligned}$$

where the second term is given in Eq. (3.8).

Finally, ϵ_t may assume a generalized error distribution (GED) with probability density function

$$f(x) = \frac{v \exp(-\frac{1}{2}|x/\lambda|^v)}{\lambda 2^{(1+1/v)} \Gamma(1/v)}, \quad -\infty < x < \infty, \quad 0 < v \leq \infty, \quad (3.9)$$

where $\Gamma(\cdot)$ is the gamma function and

$$\lambda = [2^{(-2/v)} \Gamma(1/v) / \Gamma(3/v)]^{1/2}.$$

This distribution reduces to a Gaussian distribution if $v = 2$ and it has heavy tails when $v < 2$. The conditional log likelihood function $\ell(a_{m+1}, \dots, a_T | \alpha, A_m)$ can easily be obtained.

Model Checking

For a properly specified ARCH model, the standardized residuals

$$\tilde{a}_t = \frac{a_t}{\sigma_t}$$

form a sequence of iid random variables. Therefore, one can check the adequacy of a fitted ARCH model by examining the series $\{\tilde{a}_t\}$. In particular, the Ljung–Box statistics of $\tilde{a}_t$ can be used to check the adequacy of the mean equation and that of $\tilde{a}_t^2$ can be used to test the validity of the volatility equation. The skewness, kurtosis, and quantile-to-quantile plot (i.e., QQ-plot) of $\{\tilde{a}_t\}$ can be used to check the validity of the distribution assumption. Many residual plots are available in S-Plus for model checking.

Forecasting

Forecasts of the ARCH model in Eq. (3.5) can be obtained recursively as those of an AR model. Consider an ARCH(m) model. At the forecast origin h , the 1-step ahead forecast of σ_{h+1}^2 is

$$\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \cdots + \alpha_m a_{h+1-m}^2.$$

The 2-step ahead forecast is

$$\sigma_h^2(2) = \alpha_0 + \alpha_1 \sigma_h^2(1) + \alpha_2 a_h^2 + \cdots + \alpha_m a_{h+2-m}^2,$$

and the ℓ -step ahead forecast for $\sigma_{h+\ell}^2$ is

$$\sigma_h^2(\ell) = \alpha_0 + \sum_{i=1}^m \alpha_i \sigma_h^2(\ell - i), \quad (3.10)$$

where $\sigma_h^2(\ell - i) = a_{h+\ell-i}^2$ if $\ell - i \leq 0$.

3.4.4 Some Examples

In this subsection, we illustrate ARCH modeling by considering two examples.

Example 3.1. We first apply the modeling procedure to build a simple ARCH model for the monthly log returns of Intel stock. The sample ACF and PACF of the squared returns in Figure 3.1 clearly show the existence of conditional heteroscedasticity. This is confirmed by the ARCH effect test shown in Section 3.3.1, and we proceed to identify the order of an ARCH model. The sample PACF in Figure 3.1d indicates that an ARCH(3) model might be appropriate. Consequently, we specify the model

$$r_t = \mu + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \alpha_2 a_{t-2}^2 + \alpha_3 a_{t-3}^2$$

for the monthly log returns of Intel stock. Assuming that ϵ_t are iid standard normal, we obtain the fitted model

$$r_t = 0.0171 + a_t, \quad \sigma_t^2 = 0.0120 + 0.1787a_{t-1}^2 + 0.0772a_{t-2}^2 + 0.0572a_{t-3}^2,$$

where the standard errors of the parameters are 0.0066, 0.0011, 0.0803, 0.0506, and 0.0769, respectively; see the output below. While the estimates meet the general requirement of an ARCH(3) model, the estimates of α_2 and α_3 appear to be statistically nonsignificant at the 5% level. Therefore, the model can be simplified.

S-Plus Demonstration

Output edited and % marks explanation.

```
> arch3.fit=garch(intc~1,~garch(3,0))
> summary(arch3.fit)
Call:
garch(formula.mean = intc ~ 1, formula.var = ~ garch(3, 0))

Mean Equation: intc ~ 1
Conditional Variance Equation: ~ garch(3, 0)
Conditional Distribution: gaussian
-----
Estimated Coefficients:
-----
              Value Std.Error t value Pr(>|t|)
      C 0.01713    0.006626   2.5860 0.005047 % one-sided
      A 0.01199    0.001107  10.8325 0.000000 % p-value
ARCH(1) 0.17874    0.080294   2.2260 0.013309
ARCH(2) 0.07720    0.050552   1.5271 0.063800
ARCH(3) 0.05722    0.076928   0.7438 0.228747
-----
> arch1=garch(intc~1,~garch(1,0)) % A simplified model
> summary(arch1)
Call:
garch(formula.mean = intc ~ 1, formula.var = ~ garch(1,0))

Mean Equation: intc ~ 1
Conditional Variance Equation: ~ garch(1, 0)
Conditional Distribution: gaussian
-----
Estimated Coefficients:
-----
              Value Std.Error t value  Pr(>|t|)
      C 0.01741    0.006231   2.794 2.737e-03
      A 0.01258    0.001246  10.091 0.000e+00
ARCH(1) 0.35258    0.088515   3.983 4.094e-05
-----
> stdresi=arch1$residuals/arch1$sigma.t % Standardized
> autocorTest(stdresi,lag=10)           % residuals
```

Null Hypothesis: no autocorrelation

Test Statistics:

Test Stat 13.7820 p.value 0.1832

Dist. under Null: chi-square with 10 degrees of freedom

```
> archTest(stdresi,lag=10) % ARCH test for residuals
```

Null Hypothesis: no ARCH effects

Test Statistics:

Test Stat 11.3793 p.value 0.3287

Dist. under Null: chi-square with 10 degrees of freedom

```
> arch1$asympt.sd % Obtain unconditional variance
```

```
[1] 0.1393796
```

```
> plot(arch1) % Obtain various plots, including the
               % fitted volatility series.
```

Dropping the two nonsignificant parameters, we obtain the model

$$r_t = 0.0174 + a_t, \quad \sigma_t^2 = 0.0126 + 0.3526a_{t-1}^2, \quad (3.11)$$

where the standard errors of the parameters are 0.0062, 0.0012, and 0.0885, respectively. All the estimates are highly significant. Figure 3.4 shows the standardized residuals $\{\tilde{a}_t\}$ and the sample ACF of some functions of $\{\tilde{a}_t\}$. The Ljung–Box statistics of standardized residuals give $Q(10) = 13.78$ with p -value 0.18 and those of $\{\tilde{a}_t^2\}$ give $Q(10) = 11.38$ with p -value 0.33. See the output. Consequently, the ARCH(1) model in Eq. (3.11) is adequate for describing the conditional heteroscedasticity of the data at the 5% significance level.

The ARCH(1) model in Eq. (3.11) has some interesting properties. First, the expected monthly log return for Intel stock is about 1.74%, which is remarkable, especially since the sample includes the period after the Internet bubble. Second, $\hat{\alpha}_1^2 = 0.353^2 < \frac{1}{3}$ so that the unconditional fourth moment of the monthly log return of Intel stock exists. Third, the unconditional standard deviation of r_t is $\sqrt{0.0126/(1-0.352)} = 0.1394$. Finally, the ARCH(1) model can be used to predict the monthly volatility of Intel stock returns.

t Innovation

For comparison, we also fit an ARCH(1) model with Student- t innovations to the series. The resulting model is

$$r_t = 0.0221 + a_t, \quad \sigma_t^2 = 0.0134 + 0.2492a_{t-1}^2, \quad (3.12)$$

where standard errors of the parameters are 0.0060, 0.0020, and 0.1156, respectively. The estimated degrees of freedom is 6.16 with standard error 1.65. All the estimates are significant at the 5% level, but the t -ratio of $\hat{\alpha}_1$ is only 2.16. The unconditional standard deviation of a_t is $\sqrt{0.0134/(1-0.2492)} = 0.1336$, which

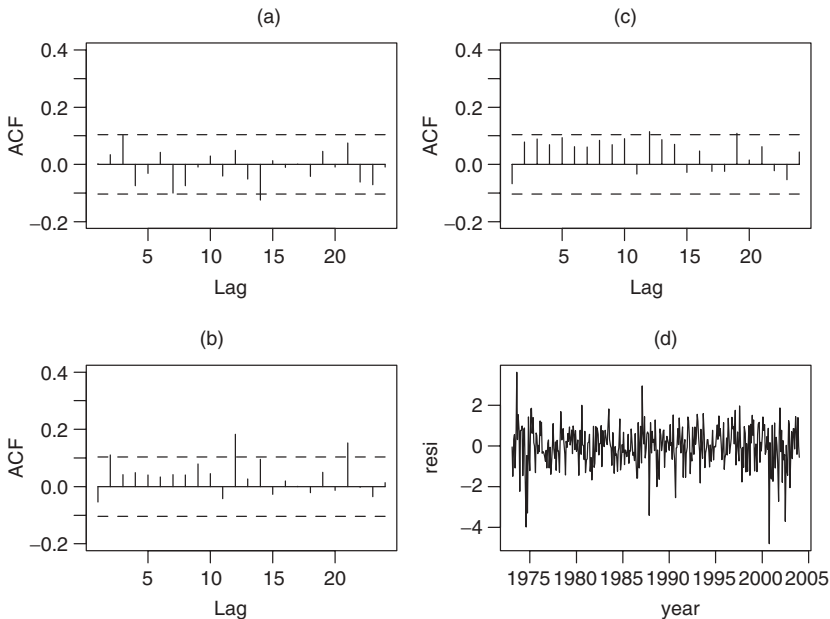


Figure 3.4. Model checking statistics of the Gaussian ARCH(1) model in Eq. (3.11) for the monthly log returns of Intel stock from January 1973 to December 2003: parts (a), (b), and (c) show the sample ACF of the standardized residuals, their squared series, and absolute series, respectively; part (d) is the time plot of standardized residuals.

is close to that obtained under normality. The Ljung–Box statistics of the standardized residuals give $Q(12) = 16.1$ with p -value 0.19, confirming that the mean equation is adequate. However, the Ljung–Box statistics for the squared standardized residuals show $Q(12) = 29.91$ with p -value 0.0029. The volatility equation is inadequate at the 5% level. Further analysis shows that the Lagrange multiplier test gives $Q(10) = 13.07$ with p -value 0.22. The inadequacy in the volatility equation is due to some higher order serial dependence in the squared standardized residuals.

Comparing models (3.11) and (3.12), we see that (a) using a heavy-tailed distribution for ϵ_t reduces the ARCH effect, and (b) the difference between the two models is small for this particular instance. Finally, a more appropriate conditional heteroscedastic model for this data set is a GARCH(1,1) model, which is discussed in the next section.

S-Plus Demonstration

With t innovations.

```
> arch1t=garch(intc~1,~garch(1,0),cond.dist='t')
> summary(arch1t)
Call:
garch(formula.mean = intc ~ 1, formula.var = ~ garch(1, 0),
      cond.dist = "t")
```

```

Mean Equation: intc ~ 1
Conditional Variance Equation: ~ garch(1, 0)
Conditional Distribution: t
  with estimated parameter 6.159751 and standard error 1.647094
Estimated Coefficients:

```

```

-----
          Value Std.Error t value  Pr(>|t|)
      C 0.02213   0.006010   3.681 1.333e-04
      A 0.01338   0.001965   6.809 2.001e-11
ARCH(1) 0.24916   0.115574   2.156 1.587e-02
-----

```

```

AIC(4) = -477.9073, BIC(4) = -462.2317

```

```

Ljung-Box test for standardized residuals:
-----

```

```

Statistic P-value Chi^2-d.f.
      16.1   0.1868          12

```

```

Ljung-Box test for squared standardized residuals:
-----

```

```

Statistic P-value Chi^2-d.f.
      29.91 0.002882          12

```

Remark. In S-Plus, the command `garch` allows for several conditional distributions. They are specified by `cond.dist = 't'` or `'ged'`. The default is Gaussian. $\square$

Example 3.2. Consider the percentage changes of the exchange rate between mark and dollar in 10-minute intervals. The data are shown in Figure 3.2a. As shown in Figure 3.3a, the series has no serial correlations. However, the sample PACF of the squared series a_t^2 shows some big spikes, especially at lags 1 and 3. There are some large PACF at higher lags, but the lower order lags tend to be more important. Following the procedure discussed in the previous subsection, we specify an ARCH(3) model for the series. Using the conditional Gaussian likelihood function, we obtain the fitted model $r_t = 0.0018 + \sigma_t \epsilon_t$ and

$$\sigma_t^2 = 0.22 \times 10^{-2} + 0.322a_{t-1}^2 + 0.074a_{t-2}^2 + 0.093a_{t-3}^2,$$

where all the estimates in the volatility equation are statistically significant at the 5% significance level. The standard errors of the volatility parameters are 0.47×10^{-6} , 0.017, 0.016, and 0.014, respectively. Model checking, using the standardized residual $\tilde{a}_t$, indicates that the model is adequate.

3.5 THE GARCH MODEL

Although the ARCH model is simple, it often requires many parameters to adequately describe the volatility process of an asset return. For instance, consider the

monthly excess returns of the S&P 500 index of Example 3.3 below. An ARCH(9) model is needed for the volatility process. Some alternative models must be sought. Bollerslev (1986) proposes a useful extension known as the generalized ARCH (GARCH) model. For a log return series r_t , let $a_t = r_t - \mu_t$ be the innovation at time t . Then a_t follows a GARCH(m, s) model if

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2, \quad (3.13)$$

where again $\{\epsilon_t\}$ is a sequence of iid random variables with mean 0 and variance 1.0, $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$. Here it is understood that $\alpha_i = 0$ for $i > m$ and $\beta_j = 0$ for $j > s$. The latter constraint on $\alpha_i + \beta_i$ implies that the unconditional variance of a_t is finite, whereas its conditional variance σ_t^2 evolves over time. As before, ϵ_t is often assumed to be a standard normal or standardized Student- t distribution or generalized error distribution. Equation (3.13) reduces to a pure ARCH(m) model if $s = 0$. The α_i and β_j are referred to as ARCH and GARCH parameters, respectively.

To understand properties of GARCH models, it is informative to use the following representation. Let $\eta_t = a_t^2 - \sigma_t^2$ so that $\sigma_t^2 = a_t^2 - \eta_t$. By plugging $\sigma_{t-i}^2 = a_{t-i}^2 - \eta_{t-i}$ ($i = 0, \dots, s$) into Eq. (3.13), we can rewrite the GARCH model as

$$a_t^2 = \alpha_0 + \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^s \beta_j \eta_{t-j}. \quad (3.14)$$

It is easy to check that $\{\eta_t\}$ is a martingale difference series (i.e., $E(\eta_t) = 0$ and $\text{cov}(\eta_t, \eta_{t-j}) = 0$ for $j \geq 1$). However, $\{\eta_t\}$ in general is not an iid sequence. Equation (3.14) is an ARMA form for the squared series a_t^2 . Thus, a GARCH model can be regarded as an application of the ARMA idea to the squared series a_t^2 . Using the unconditional mean of an ARMA model, we have

$$E(a_t^2) = \frac{\alpha_0}{1 - \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i)}$$

provided that the denominator of the prior fraction is positive.

The strengths and weaknesses of GARCH models can easily be seen by focusing on the simplest GARCH(1,1) model with

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad 0 \leq \alpha_1, \beta_1 \leq 1, (\alpha_1 + \beta_1) < 1. \quad (3.15)$$

First, a large a_{t-1}^2 or σ_{t-1}^2 gives rise to a large σ_t^2 . This means that a large a_{t-1}^2 tends to be followed by another large a_t^2 , generating, again, the well-known behavior of volatility clustering in financial time series. Second, it can be shown that if $1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2 > 0$, then

$$\frac{E(a_t^4)}{[E(a_t^2)]^2} = \frac{3[1 - (\alpha_1 + \beta_1)^2]}{1 - (\alpha_1 + \beta_1)^2 - 2\alpha_1^2} > 3.$$

Consequently, similar to ARCH models, the tail distribution of a GARCH(1,1) process is heavier than that of a normal distribution. Third, the model provides a simple parametric function that can be used to describe the volatility evolution.

Forecasts of a GARCH model can be obtained using methods similar to those of an ARMA model. Consider the GARCH(1,1) model in Eq. (3.15) and assume that the forecast origin is h . For 1-step ahead forecast, we have

$$\sigma_{h+1}^2 = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2,$$

where a_h and σ_h^2 are known at the time index h . Therefore, the 1-step ahead forecast is

$$\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2.$$

For multistep ahead forecasts, we use $a_t^2 = \sigma_t^2 \epsilon_t^2$ and rewrite the volatility equation in Eq. (3.15) as

$$\sigma_{t+1}^2 = \alpha_0 + (\alpha_1 + \beta_1) \sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1).$$

When $t = h + 1$, the equation becomes

$$\sigma_{h+2}^2 = \alpha_0 + (\alpha_1 + \beta_1) \sigma_{h+1}^2 + \alpha_1 \sigma_{h+1}^2 (\epsilon_{h+1}^2 - 1).$$

Since $E(\epsilon_{h+1}^2 - 1 | F_h) = 0$, the 2-step ahead volatility forecast at the forecast origin h satisfies the equation

$$\sigma_h^2(2) = \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2(1).$$

In general, we have

$$\sigma_h^2(\ell) = \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2(\ell - 1), \quad \ell > 1. \quad (3.16)$$

This result is exactly the same as that of an ARMA(1,1) model with AR polynomial $1 - (\alpha_1 + \beta_1)B$. By repeated substitutions in Eq. (3.16), we obtain that the ℓ -step ahead forecast can be written as

$$\sigma_h^2(\ell) = \frac{\alpha_0 [1 - (\alpha_1 + \beta_1)^{\ell-1}]}{1 - \alpha_1 - \beta_1} + (\alpha_1 + \beta_1)^{\ell-1} \sigma_h^2(1).$$

Therefore,

$$\sigma_h^2(\ell) \rightarrow \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \text{as } \ell \rightarrow \infty$$

provided that $\alpha_1 + \beta_1 < 1$. Consequently, the multistep ahead volatility forecasts of a GARCH(1,1) model converge to the unconditional variance of a_t as the forecast horizon increases to infinity provided that $\text{Var}(a_t)$ exists.

The literature on GARCH models is enormous; see Bollerslev, Chou, and Kroner (1992), Bollerslev, Engle, and Nelson (1994), and the references therein. The model encounters the same weaknesses as the ARCH model. For instance, it responds equally to positive and negative shocks. In addition, recent empirical studies of high-frequency financial time series indicate that the tail behavior of GARCH models remains too short even with standardized Student- t innovations. For further information about kurtosis of GARCH models, see Section 3.16.

3.5.1 An Illustrative Example

The modeling procedure of ARCH models can also be used to build a GARCH model. However, specifying the order of a GARCH model is not easy. Only lower order GARCH models are used in most applications, say, GARCH(1,1), GARCH(2,1), and GARCH(1,2) models. The conditional maximum likelihood method continues to apply provided that the starting values of the volatility $\{\sigma_t^2\}$ are assumed to be known. Consider, for instance, a GARCH(1,1) model. If σ_1^2 is treated as fixed, then σ_t^2 can be computed recursively for a GARCH(1,1) model. In some applications, the sample variance of a_t serves as a good starting value of σ_1^2 . The fitted model can be checked by using the standardized residual $\tilde{a}_t = a_t/\sigma_t$ and its squared process.

Example 3.3. In this example, we consider the monthly excess returns of the S&P 500 index starting from 1926 for 792 observations. The series is shown in Figure 3.5. Denote the excess return series by r_t . Figure 3.6 shows the sample ACF of r_t and the sample PACF of r_t^2 . The r_t series has some serial correlations at lags 1 and 3, but the key feature is that the PACF of r_t^2 shows strong linear dependence. If an MA(3) model is entertained, we obtain

$$r_t = 0.0062 + a_t + 0.0944a_{t-1} - 0.1407a_{t-3}, \quad \hat{\sigma}_a = 0.0576$$

for the series, where all of the coefficients are significant at the 5% level. However, for simplicity, we use instead an AR(3) model

$$r_t = \phi_1 r_{t-1} + \phi_2 r_{t-2} + \phi_3 r_{t-3} + \beta_0 + a_t.$$

The fitted AR(3) model, under the normality assumption, is

$$r_t = 0.088r_{t-1} - 0.023r_{t-2} - 0.123r_{t-3} + 0.0066 + a_t, \quad \hat{\sigma}_a^2 = 0.00333. \quad (3.17)$$

For the GARCH effects, we use the GARCH(1,1) model

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + \alpha_1 a_{t-1}^2.$$

A joint estimation of the AR(3)–GARCH(1,1) model gives

$$\begin{aligned} r_t &= 0.0078 + 0.032r_{t-1} - 0.029r_{t-2} - 0.008r_{t-3} + a_t, \\ \sigma_t^2 &= 0.000084 + 0.1213a_{t-1}^2 + 0.8523\sigma_{t-1}^2. \end{aligned}$$

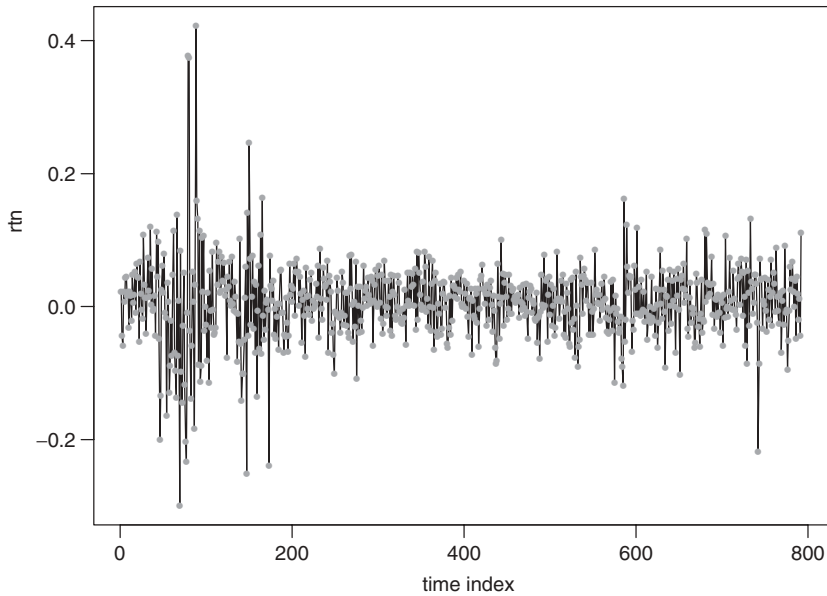


Figure 3.5. Time series plot of the monthly excess returns of the S&P 500 index.

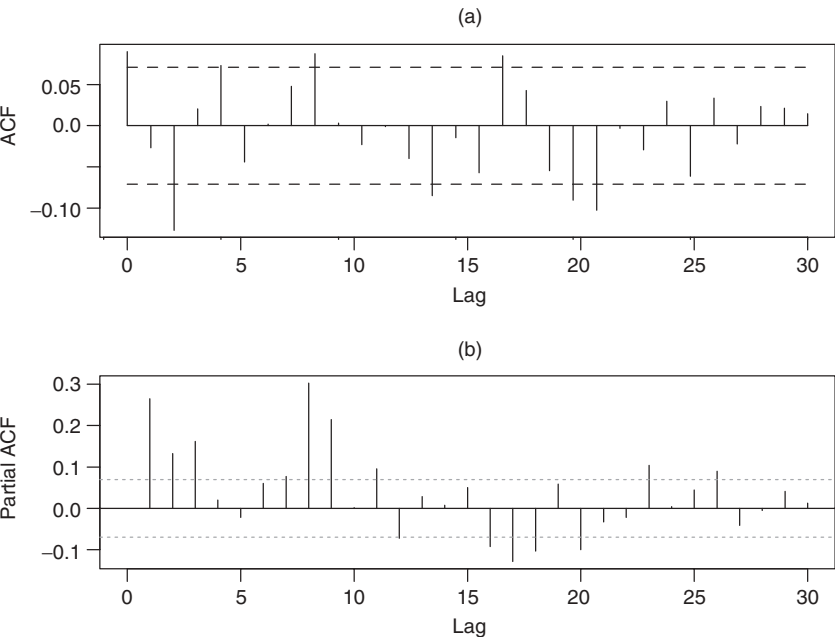


Figure 3.6. (a) Sample ACF of the monthly excess returns of the S&P 500 index and (b) sample PACF of the squared monthly excess returns.

From the volatility equation, the implied unconditional variance of a_t is

$$\frac{0.000084}{1 - 0.8523 - 0.1213} = 0.00317,$$

which is close to that of Eq. (3.17). However, t -ratios of the parameters in the mean equation suggest that all three AR coefficients are insignificant at the 5% level. Therefore, we refine the model by dropping all AR parameters. The refined model is

$$r_t = 0.0076 + a_t, \quad \sigma_t^2 = 0.000086 + 0.1216a_{t-1}^2 + 0.8511\sigma_{t-1}^2. \quad (3.18)$$

The standard error of the constant in the mean equation is 0.0015, whereas those of the parameters in the volatility equation are 0.000024, 0.0197, and 0.0190, respectively. The unconditional variance of a_t is $0.000086/(1 - 0.8511 - 0.1216) = 0.00314$. This is a simple stationary GARCH(1,1) model. Figure 3.7 shows the estimated volatility process, σ_t , and the standardized shocks $\tilde{a}_t = a_t/\sigma_t$ for the GARCH(1,1) model in Eq. (3.18). The $\tilde{a}_t$ series appears to be a white noise process. Figure 3.8 provides the sample ACF of the standardized residuals $\tilde{a}_t$ and

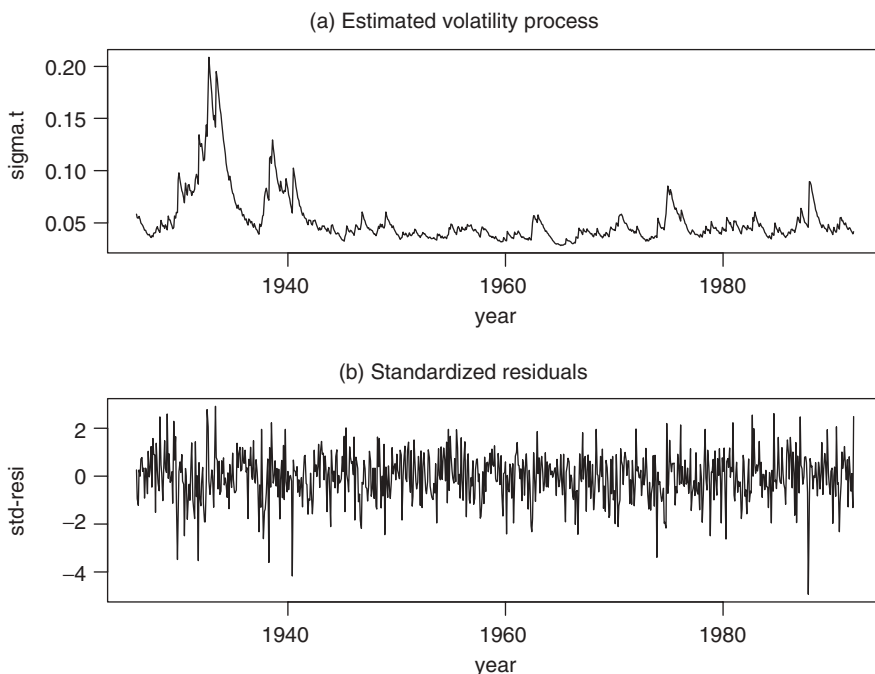


Figure 3.7. (a) Time series plot of estimated volatility (σ_t) for the monthly excess returns of the S&P 500 index and (b) the standardized shocks of the monthly excess returns of the S&P 500 index. Both plots are based on the GARCH(1,1) model in Eq. (3.18).

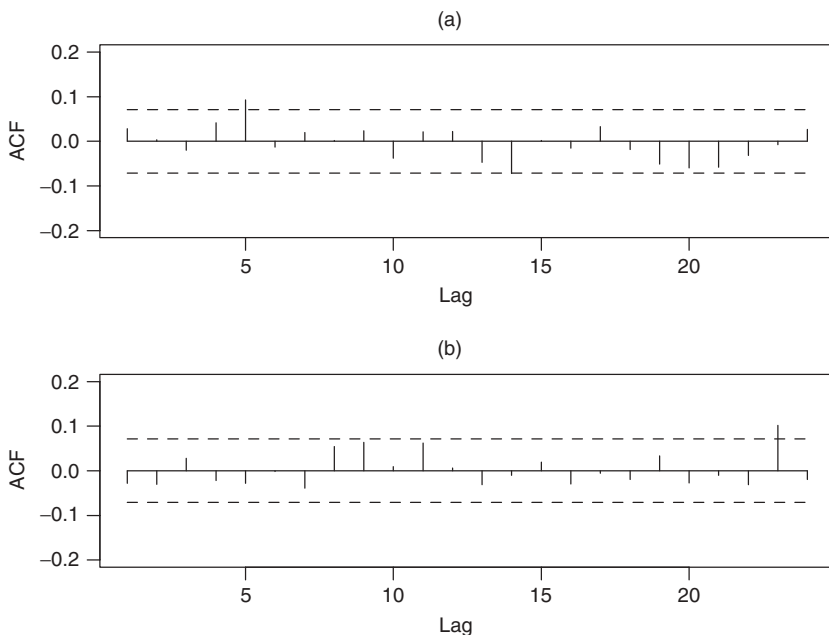


Figure 3.8. Model checking of the GARCH(1,1) model in Eq. (3.18) for monthly excess returns of the S&P 500 index: (a) sample ACF of standardized residuals and (b) sample ACF of the squared standardized residuals.

the squared process $\tilde{a}_t^2$. These ACFs fail to suggest any significant serial correlations or conditional heteroscedasticity in the standardized residual series. More specifically, we have $Q(12) = 11.99(0.45)$ and $Q(24) = 28.52(0.24)$ for $\tilde{a}_t$, and $Q(12) = 13.11(0.36)$ and $Q(24) = 26.45(0.33)$ for $\tilde{a}_t^2$, where the number in parentheses is the p -value of the test statistic. Thus, the model appears to be adequate in describing the linear dependence in the return and volatility series. Note that the fitted model shows $\hat{\alpha}_1 + \hat{\beta}_1 = 0.9772$, which is close to 1. This phenomenon is commonly observed in practice and it leads to imposing the constraint $\alpha_1 + \beta_1 = 1$ in a GARCH(1,1) model, resulting in an integrated GARCH (or IGARCH) model; see Section 3.6.

Finally, to forecast the volatility of monthly excess returns of the S&P 500 index, we can use the volatility equation in Eq. (3.18). For instance, at the forecast origin h , we have $\sigma_{h+1}^2 = 0.000086 + 0.1216a_h^2 + 0.8511\sigma_h^2$. The 1-step ahead forecast is then

$$\sigma_h^2(1) = 0.000086 + 0.1216a_h^2 + 0.8511\sigma_h^2,$$

where a_h is the residual of the mean equation at time h and σ_h is obtained from the volatility equation. The starting value σ_0^2 is fixed at either zero or the unconditional variance of a_t . For multistep ahead forecasts, we use the recursive formula in Eq. (3.16). Table 3.1 shows some mean and volatility forecasts for the monthly

Table 3.1. Volatility Forecasts for the Monthly Excess Returns of the S&P 500 Index^a

Horizon	1	2	3	4	5	∞
Return	0.0076	0.0076	0.0076	0.0076	0.0076	0.0076
Volatility	0.0536	0.0537	0.0537	0.0538	0.0538	0.0560

^aThe forecast origin is $h = 792$, which corresponds to December 1991. Here volatility denotes conditional standard deviation.

excess return of the S&P 500 index with forecast origin $h = 792$ based on the GARCH(1,1) model in Eq. (3.18).

Some S-Plus Commands Used in Example 3.3

```
> fit=garch(sp~ar(3),~garch(1,1))
> summary(fit)
> fit=garch(sp~1,~garch(1,1))
> summary(fit)
> names(fit)
[1] "residuals" "sigma.t"      "df.residual" "coef" "model"
[6] "cond.dist"  "likelihood" "opt.index"   "cov"
[10] "prediction" "call"        "asypm.sd"    "series"
> % Next, compute the standardized residuals
> stdresi=fit$residuals/fit$sigma.t
> autocorTest(stdresi,lag=24)
> autocorTest(stdresi^2,lag=24)
> predict(fit,5) % Compute predictions
```

Note that in the prior commands the volatility series σ_t is stored in `fit$sigma.t` and the residual series of the returns in `fit$residuals`.

t Innovation

Assuming that ϵ_t follows a standardized Student- t distribution with 5 degrees of freedom, we reestimate the GARCH(1,1) model and obtain

$$r_t = 0.0085 + a_t, \quad \sigma_t^2 = 0.00012 + 0.1121a_{t-1}^2 + 0.8432\sigma_{t-1}^2, \quad (3.19)$$

where the standard errors of the parameters are 0.0015, 0.51×10^{-4} , 0.0296, and 0.0371, respectively. This model is essentially an IGARCH(1,1) model as $\hat{\alpha}_1 + \hat{\beta}_1 \approx 0.95$, which is close to 1. The Ljung–Box statistics of the standardized residuals give $Q(10) = 11.38$ with p -value 0.33 and those of the $\{\hat{a}_t^2\}$ series give $Q(10) = 10.48$ with p -value 0.40. Thus, the fitted GARCH(1,1) model with Student- t distribution is adequate.

S-Plus Commands Used

```
> fit1 = garch(sp~1,~garch(1,1),cond.dist='t',cond.par=5,
+ cond.est=F)
> summary(fit1)
> stres1=fit1$residuals/fit1$sigma.t
> autocorTest(stres1,lag=10)
> autocorTest(stres1^2,lag=10)
```

Estimation of Degrees of Freedom

If we further extend the GARCH(1,1) model by estimating the degrees of freedom of the Student- t distribution used, we obtain the model

$$r_t = 0.0085 + a_t, \quad \sigma_t^2 = 0.00012 + 0.1121a_{t-1}^2 + 0.8432\sigma_{t-1}^2, \quad (3.20)$$

where the estimated degrees of freedom is 7.02. Standard errors of the estimates in Eq. (3.20) are close to those in Eq. (3.19). The standard error of the estimated degrees of freedom is 1.78. Consequently, we cannot reject the hypothesis of using a standardized Student- t distribution with 5 degrees of freedom at the 5% significance level.

S-Plus Commands Used

```
> fit2 = garch(sp~1, ~garch(1,1), cond.dist='t')
> summary(fit2)
```

3.5.2 Forecasting Evaluation

Since the volatility of an asset return is not directly observable, comparing the forecasting performance of different volatility models is a challenge to data analysts. In the literature, some researchers use out-of-sample forecasts and compare the volatility forecasts $\sigma_h^2(\ell)$ with the shock $a_{h+\ell}^2$ in the forecasting sample to assess the forecasting performance of a volatility model. This approach often finds a low correlation coefficient between $a_{h+\ell}^2$ and $\sigma_h^2(\ell)$, that is, low R^2 . However, such a finding is not surprising because $a_{h+\ell}^2$ alone is not an adequate measure of the volatility at time index $h + \ell$. Consider the 1-step ahead forecasts. From a statistical point of view, $E(a_{h+1}^2 | F_h) = \sigma_{h+1}^2$ so that a_{h+1}^2 is a consistent estimate of σ_{h+1}^2 . But it is not an accurate estimate of σ_{h+1}^2 because a single observation of a random variable with a known mean value cannot provide an accurate estimate of its variance. Consequently, such an approach to evaluate forecasting performance of volatility models is strictly speaking not proper. For more information concerning forecasting evaluation of GARCH models, readers are referred to Andersen and Bollerslev (1998).

3.5.3 A Two-Pass Estimation Method

Based on Eq. (3.14), a two-pass estimation method can be used to estimate GARCH models. First, ignoring any ARCH effects, one estimates the mean equation of a return series using the methods discussed in Chapter 2 (e.g., maximum likelihood method). Denote the residual series by a_t . Second, treating $\{a_t^2\}$ as an observed time series, one applies the maximum likelihood method to estimate parameters of Eq. (3.14). Denote the AR and MA coefficient estimates by $\hat{\phi}_i$ and $\hat{\theta}_i$. The GARCH estimates are obtained as $\hat{\beta}_i = \hat{\theta}_i$ and $\hat{\alpha}_i = \hat{\phi}_i - \hat{\theta}_i$. Obviously, such estimates are approximations to the true parameters and their statistical properties have not been rigorously investigated. However, limited experience shows that this simple approach often provides good approximations, especially when the sample size

is moderate or large. For instance, consider the monthly excess return series of the S&P 500 index of Example 3.3. Using the conditional MLE method in SCA, we obtain the model

$$r_t = 0.0061 + a_t, \quad a_t^2 = 0.00014 + 0.9583a_{t-1}^2 + \eta_t - 0.8456\eta_{t-1},$$

where all estimates are significantly different from zero at the 5% level. From the estimates, we have $\hat{\beta}_1 = 0.8456$ and $\hat{\alpha}_1 = 0.9583 - 0.8456 = 0.1127$. These approximate estimates are very close to those in Eq. (3.18) or (3.20). Furthermore, the fitted volatility series of the two-pass method is very close to that of Figure 3.7a.

3.6 THE INTEGRATED GARCH MODEL

If the AR polynomial of the GARCH representation in Eq. (3.14) has a unit root, then we have an IGARCH model. Thus, IGARCH models are unit-root GARCH models. Similar to ARIMA models, a key feature of IGARCH models is that the impact of past squared shocks $\eta_{t-i} = a_{t-i}^2 - \sigma_{t-i}^2$ for $i > 0$ on a_t^2 is persistent.

An IGARCH(1,1) model can be written as

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2,$$

where $\{\epsilon_t\}$ is defined as before and $1 > \beta_1 > 0$. For the monthly excess returns of the S&P 500 index, an estimated IGARCH(1,1) model is

$$r_t = 0.0067 + a_t, \quad a_t = \sigma_t \epsilon_t, \\ \sigma_t^2 = 0.000119 + 0.8059 \sigma_{t-1}^2 + 0.1941 a_{t-1}^2,$$

where the standard errors of the estimates in the volatility equation are 0.0017, 0.000013, and 0.0144, respectively. The parameter estimates are close to those of the GARCH(1,1) model shown before, but there is a major difference between the two models. The unconditional variance of a_t , hence that of r_t , is not defined under the above IGARCH(1,1) model. This seems hard to justify for an excess return series. From a theoretical point of view, the IGARCH phenomenon might be caused by occasional level shifts in volatility. The actual cause of persistence in volatility deserves a careful investigation.

When $\alpha_1 + \beta_1 = 1$, repeated substitutions in Eq. (3.16) give

$$\sigma_h^2(\ell) = \sigma_h^2(1) + (\ell - 1)\alpha_0, \quad \ell \geq 1, \quad (3.21)$$

where h is the forecast origin. Consequently, the effect of $\sigma_h^2(1)$ on future volatilities is also persistent, and the volatility forecasts form a straight line with slope α_0 . Nelson (1990) studies some probability properties of the volatility process σ_t^2 under an IGARCH model. The process σ_t^2 is a martingale for which some nice results are available in the literature. Under certain conditions, the volatility process is strictly stationary, but not weakly stationary because it does not have the first two moments.

The case of $\alpha_0 = 0$ is of particular interest in studying the IGARCH(1,1) model. In this case, the volatility forecasts are simply $\sigma_h^2(1)$ for all forecast horizons; see Eq. (3.21). This special IGARCH(1,1) model is the volatility model used in RiskMetrics, which is an approach for calculating value at risk; see Chapter 7. The model is also an exponential smoothing model for the $\{a_t^2\}$ series. To see this, rewrite the model as

$$\begin{aligned}\sigma_t^2 &= (1 - \beta_1)a_{t-1}^2 + \beta_1\sigma_{t-1}^2 \\ &= (1 - \beta_1)a_{t-1}^2 + \beta_1[(1 - \beta)a_{t-2}^2 + \beta_1\sigma_{t-2}^2] \\ &= (1 - \beta_1)a_{t-1}^2 + (1 - \beta_1)\beta_1a_{t-2}^2 + \beta_1^2\sigma_{t-2}^2\end{aligned}$$

By repeated substitutions, we have

$$\sigma_t^2 = (1 - \beta_1)(a_{t-1}^2 + \beta_1a_{t-2}^2 + \beta_1^2a_{t-3}^2 + \cdots),$$

which is the well-known exponential smoothing formation with β_1 being the discounting factor. Exponential smoothing methods can thus be used to estimate such an IGARCH(1,1) model.

3.7 THE GARCH-M MODEL

In finance, the return of a security may depend on its volatility. To model such a phenomenon, one may consider the GARCH-M model, where “M” stands for GARCH *in the mean*. A simple GARCH(1,1)-M model can be written as

$$\begin{aligned}r_t &= \mu + c\sigma_t^2 + a_t, \quad a_t = \sigma_t\epsilon_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1a_{t-1}^2 + \beta_1\sigma_{t-1}^2,\end{aligned}\tag{3.22}$$

where μ and c are constants. The parameter c is called the risk premium parameter. A positive c indicates that the return is positively related to its volatility. Other specifications of risk premium have also been used in the literature, including $r_t = \mu + c\sigma_t + a_t$ and $r_t = \mu + c\ln(\sigma_t^2) + a_t$.

The formulation of the GARCH-M model in Eq. (3.22) implies that there are serial correlations in the return series r_t . These serial correlations are introduced by those in the volatility process $\{\sigma_t^2\}$. The existence of risk premium is, therefore, another reason that some historical stock returns have serial correlations.

For illustration, we consider a GARCH(1,1)-M model with Gaussian innovations for the monthly excess returns of the S&P 500 index from January 1926 to December 1991. The fitted model is

$$r_t = 0.0055 + 1.09\sigma_t^2 + a_t, \quad \sigma_t^2 = 8.76 \times 10^{-5} + 0.123a_{t-1}^2 + 0.849\sigma_{t-1}^2,$$

where the standard errors for the two parameters in the mean equation are 0.0023 and 0.818, respectively, and those for the parameters in the volatility equation are 2.51×10^{-5} , 0.0205, and 0.0196, respectively. The estimated risk premium for the index return is positive but is not statistically significant at the 5% level. Here the result is obtained using S-Plus. Other forms of GARCH-M specification in S-Plus are given in Table 3.2. The idea of risk premium applies to other GARCH models.

**Table 3.2. GARCH-M Models
Allowed in S-Plus^a**

$g(\sigma_t)$	Command
σ_t^2	var.in.mean
σ_t	sd.in.mean
$\ln(\sigma_t^2)$	logvar.in.mean

^aThe mean equation is $r_t = \mu + c g(\sigma_t) + a_t$.

S-Plus Demonstration

```
> sp.fit = garch(sp~1+var.in.mean, ~garch(1,1))
> summary(sp.fit)
```

3.8 THE EXPONENTIAL GARCH MODEL

To overcome some weaknesses of the GARCH model in handling financial time series, Nelson (1991) proposes the exponential GARCH (EGARCH) model. In particular, to allow for asymmetric effects between positive and negative asset returns, he considers the weighted innovation

$$g(\epsilon_t) = \theta \epsilon_t + \gamma [|\epsilon_t| - E(|\epsilon_t|)], \quad (3.23)$$

where θ and γ are real constants. Both ϵ_t and $|\epsilon_t| - E(|\epsilon_t|)$ are zero-mean iid sequences with continuous distributions. Therefore, $E[g(\epsilon_t)] = 0$. The asymmetry of $g(\epsilon_t)$ can easily be seen by rewriting it as

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0. \end{cases}$$

Remark. For the standard Gaussian random variable ϵ_t , $E(|\epsilon_t|) = \sqrt{2/\pi}$. For the standardized Student- t distribution in Eq. (3.7), we have

$$E(|\epsilon_t|) = \frac{2\sqrt{v-2}\Gamma((v+1)/2)}{(v-1)\Gamma(v/2)\sqrt{\pi}}. \quad \square$$

An EGARCH(m, s) model can be written as

$$a_t = \sigma_t \epsilon_t, \quad \ln(\sigma_t^2) = \alpha_0 + \frac{1 + \beta_1 B + \cdots + \beta_{s-1} B^{s-1}}{1 - \alpha_1 B - \cdots - \alpha_m B^m} g(\epsilon_{t-1}), \quad (3.24)$$

where α_0 is a constant, B is the back-shift (or lag) operator such that $Bg(\epsilon_t) = g(\epsilon_{t-1})$, and $1 + \beta_1 B + \cdots + \beta_{s-1} B^{s-1}$ and $1 - \alpha_1 B - \cdots - \alpha_m B^m$ are polynomials with zeros outside the unit circle and have no common factors. By outside the unit circle, we mean that absolute values of the zeros are greater than 1. Again, Eq. (3.24) uses the usual ARMA parameterization to describe the evolution of the conditional variance of a_t . Based on this representation, some properties of the EGARCH model can be obtained in a similar manner as those of the GARCH

model. For instance, the unconditional mean of $\ln(\sigma_t^2)$ is α_0 . However, the model differs from the GARCH model in several ways. First, it uses logged conditional variance to relax the positiveness constraint of model coefficients. Second, the use of $g(\epsilon_t)$ enables the model to respond asymmetrically to positive and negative lagged values of a_t . Some additional properties of the EGARCH model can be found in Nelson (1991).

To better understand the EGARCH model, let us consider the simple model with order (1,1):

$$a_t = \sigma_t \epsilon_t, \quad (1 - \alpha B) \ln(\sigma_t^2) = (1 - \alpha)\alpha_0 + g(\epsilon_{t-1}), \quad (3.25)$$

where the ϵ_t are iid standard normal and the subscript of α_1 is omitted. In this case, $E(|\epsilon_t|) = \sqrt{2/\pi}$ and the model for $\ln(\sigma_t^2)$ becomes

$$(1 - \alpha B) \ln(\sigma_t^2) = \begin{cases} \alpha_* + (\gamma + \theta)\epsilon_{t-1} & \text{if } \epsilon_{t-1} \geq 0, \\ \alpha_* + (\gamma - \theta)(-\epsilon_{t-1}) & \text{if } \epsilon_{t-1} < 0, \end{cases} \quad (3.26)$$

where $\alpha_* = (1 - \alpha)\alpha_0 - \sqrt{2/\pi}\gamma$. This is a nonlinear function similar to that of the threshold autoregressive (TAR) model of Tong (1978, 1990). It suffices to say that for this simple EGARCH model the conditional variance evolves in a nonlinear manner depending on the sign of a_{t-1} . Specifically, we have

$$\sigma_t^2 = \sigma_{t-1}^{2\alpha} \exp(\alpha_*) \begin{cases} \exp\left((\gamma + \theta)\frac{a_{t-1}}{\sigma_{t-1}}\right) & \text{if } a_{t-1} \geq 0, \\ \exp\left((\gamma - \theta)\frac{|a_{t-1}|}{\sigma_{t-1}}\right) & \text{if } a_{t-1} < 0. \end{cases}$$

The coefficients $(\gamma + \theta)$ and $(\gamma - \theta)$ show the asymmetry in response to positive and negative a_{t-1} . The model is, therefore, nonlinear if $\theta \neq 0$. Since negative shocks tend to have larger impacts, we expect θ to be negative. For higher order EGARCH models, the nonlinearity becomes much more complicated. Cao and Tsay (1992) use nonlinear models, including EGARCH models, to obtain multi-step ahead volatility forecasts. We discuss nonlinearity in financial time series in Chapter 4.

3.8.1 An Alternative Model Form

An alternative form for the EGARCH(m, s) model is

$$\ln(\sigma_t^2) = \alpha_0 + \sum_{i=1}^s \alpha_i \frac{|a_{t-i}| + \gamma_i a_{t-i}}{\sigma_{t-i}} + \sum_{j=1}^m \beta_j \ln(\sigma_{t-j}^2). \quad (3.27)$$

Here a positive a_{t-i} contributes $\alpha_i(1 + \gamma_i)|\epsilon_{t-i}|$ to the log volatility, whereas a negative a_{t-i} gives $\alpha_i(1 - \gamma_i)|\epsilon_{t-i}|$, where $\epsilon_{t-i} = a_{t-i}/\sigma_{t-i}$. The γ_i parameter thus signifies the leverage effect of a_{t-i} . Again, we expect γ_i to be negative in real applications. This is the model form used in S-Plus.

3.8.2 An Illustrative Example

Nelson (1991) applies an EGARCH model to the daily excess returns of the value-weighted market index from the Center for Research in Security Prices from July 1962 to December 1987. The excess returns are obtained by removing monthly Treasury bill returns from the value-weighted index returns, assuming that the Treasury bill return was constant for each calendar day within a given month. There are 6408 observations. Denote the excess return by r_t . The model used is as follows:

$$\begin{aligned} r_t &= \phi_0 + \phi_1 r_{t-1} + c\sigma_t^2 + a_t, \\ \ln(\sigma_t^2) &= \alpha_0 + \ln(1 + wN_t) + \frac{1 + \beta B}{1 - \alpha_1 B - \alpha_2 B^2} g(\epsilon_{t-1}), \end{aligned} \tag{3.28}$$

where σ_t^2 is the conditional variance of a_t given F_{t-1} , N_t is the number of non-trading days between trading days $t - 1$ and t , α_0 and w are real parameters, $g(\epsilon_t)$ is defined in Eq. (3.23), and ϵ_t follows a generalized error distribution in Eq. (3.9). Similar to a GARCH-M model, the parameter c in Eq. (3.28) is the risk premium parameter. Table 3.3 gives the parameter estimates and their standard errors of the model. The mean equation of model (3.28) has two features that are of interest. First, it uses an AR(1) model to take care of possible serial correlation in the excess returns. Second, it uses the volatility σ_t^2 as a regressor to account for risk premium. The estimated risk premium is negative, but statistically insignificant.

3.8.3 Second Example

As another illustration, we consider the monthly log returns of IBM stock from January 1926 to December 1997 for 864 observations. An AR(1)–EGARCH(1,1) model is entertained and the fitted model is

$$r_t = 0.0105 + 0.092r_{t-1} + a_t, \quad a_t = \sigma_t \epsilon_t, \tag{3.29}$$

$$\ln(\sigma_t^2) = -5.496 + \frac{g(\epsilon_{t-1})}{1 - 0.856B}, \tag{3.30}$$

$$g(\epsilon_{t-1}) = -0.0795\epsilon_{t-1} + 0.2647 \left[|\epsilon_{t-1}| - \sqrt{2/\pi} \right],$$

Table 3.3. Estimated AR(1)–EGARCH(2,2) Model for the Daily Excess Returns of the Value-Weighted CRSP Market Index: July 1962 to December 1987

Parameter	α_0	w	γ	α_1	α_2	β
Estimate	−10.06	0.183	0.156	1.929	−0.929	−0.978
Error	0.346	0.028	0.013	0.015	0.015	0.006
Parameter	θ	ϕ_0	ϕ_1	c	v	
Estimate	−0.118	$3.5 \cdot 10^{-4}$	0.205	−3.361	1.576	
Error	0.009	$9.9 \cdot 10^{-5}$	0.012	2.026	0.032	

where $\{\epsilon_t\}$ is a sequence of independent standard Gaussian random variates. All parameter estimates are statistically significant at the 5% level. For model checking, the Ljung–Box statistics give $Q(10) = 6.31(0.71)$ and $Q(20) = 21.4(0.32)$ for the standardized residual process $\tilde{a}_t = a_t/\sigma_t$ and $Q(10) = 4.13(0.90)$ and $Q(20) = 15.93(0.66)$ for the squared process $\tilde{a}_t^2$, where again the number in parentheses denotes p -value. Therefore, there is no serial correlation or conditional heteroscedasticity in the standardized residuals of the fitted model. The prior AR(1)–EGARCH(1,1) model is adequate.

From the estimated volatility equation in (3.30) and using $\sqrt{2/\pi} \approx 0.7979$, we obtain the volatility equation as

$$\ln(\sigma_t^2) = -1.001 + 0.856 \ln(\sigma_{t-1}^2) + \begin{cases} 0.1852\epsilon_{t-1} & \text{if } \epsilon_{t-1} \geq 0, \\ -0.3442\epsilon_{t-1} & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

Taking antilog transformation, we have

$$\sigma_t^2 = \sigma_{t-1}^{2 \times 0.856} e^{-1.001} \times \begin{cases} e^{0.1852\epsilon_{t-1}} & \text{if } \epsilon_{t-1} \geq 0, \\ e^{-0.3442\epsilon_{t-1}} & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

This equation highlights the asymmetric responses in volatility to the past positive and negative shocks under an EGARCH model. For example, for a standardized shock with magnitude 2 (i.e., two standard deviations), we have

$$\frac{\sigma_t^2(\epsilon_{t-1} = -2)}{\sigma_t^2(\epsilon_{t-1} = 2)} = \frac{\exp[-0.3442 \times (-2)]}{\exp(0.1852 \times 2)} = e^{0.318} = 1.374.$$

Therefore, the impact of a negative shock of size two standard deviations is about 37.4% higher than that of a positive shock of the same size. This example clearly demonstrates the asymmetric feature of EGARCH models. In general, the bigger the shock, the larger the difference in volatility impact.

Finally, we extend the sample period to include the log returns from 1998 to 2003 so that there are 936 observations and use S-Plus to fit an EGARCH(1,1) model. The results are given below.

S-Plus Demonstration

Output edited.

```
> ibm.egarch=garch(ibmln~1,~egarch(1,1),leverage=T,
+ cond.dist='ged')

> summary(ibm.egarch)
Call:
garch(formula.mean = ibmln ~ 1, formula.var = ~ egarch(1,1),
      leverage = T,cond.dist = "ged")

Mean Equation: ibmln ~ 1
```

Conditional Variance Equation: ~ egarch(1, 1)
 Conditional Distribution: ged
 with estimated parameter 1.5003 and standard error 0.09912

 Estimated Coefficients:

	Value	Std.Error	t value	Pr(> t)
C	0.01181	0.002012	5.870	3.033e-09
A	-0.55680	0.171602	-3.245	6.088e-04
ARCH(1)	0.22025	0.052824	4.169	1.669e-05
GARCH(1)	0.92910	0.026743	34.742	0.000e+00
LEV(1)	-0.26400	0.126096	-2.094	1.828e-02

 Ljung-Box test for standardized residuals:

Statistic	P-value	Chi^2-d.f.
17.87	0.1195	12

Ljung-Box test for squared standardized residuals:

Statistic	P-value	Chi^2-d.f.
6.723	0.8754	12

The fitted GARCH(1,1) model is

$$r_t = 0.0118 + a_t, \quad a_t = \sigma_t \epsilon_t, \quad (3.31)$$

$$\ln(\sigma_t^2) = -0.557 + 0.220 \frac{|a_{t-1}| - 0.264a_{t-1}}{\sigma_{t-1}} + 0.929 \ln(\sigma_{t-1}^2),$$

where ϵ_t follows a GED distribution with parameter 1.5. This model is adequate based on the Ljung–Box statistics of the standardized residual series and its squared process. As expected, the output shows that the estimated leverage effect is negative and is statistically significant at the 5% level with t -ratio -2.094 .

3.8.4 Forecasting Using an EGARCH Model

We use the EGARCH(1,1) model to illustrate multistep ahead forecasts of EGARCH models, assuming that the model parameters are known and the innovations are standard Gaussian. For such a model, we have

$$\ln(\sigma_t^2) = (1 - \alpha_1)\alpha_0 + \alpha_1 \ln(\sigma_{t-1}^2) + g(\epsilon_{t-1}),$$

$$g(\epsilon_{t-1}) = \theta\epsilon_{t-1} + \gamma(|\epsilon_{t-1}| - \sqrt{2/\pi}).$$

Taking exponentials, the model becomes

$$\sigma_t^2 = \sigma_{t-1}^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_{t-1})], \quad (3.32)$$

$$g(\epsilon_{t-1}) = \theta\epsilon_{t-1} + \gamma(|\epsilon_{t-1}| - \sqrt{2/\pi}).$$

Let h be the forecast origin. For the 1-step ahead forecast, we have

$$\sigma_{h+1}^2 = \sigma_h^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_h)],$$

where all of the quantities on the right-hand side are known. Thus, the 1-step ahead volatility forecast at the forecast origin h is simply $\hat{\sigma}_h^2(1) = \sigma_{h+1}^2$ given earlier. For the 2-step ahead forecast, Eq. (3.32) gives

$$\sigma_{h+2}^2 = \sigma_{h+1}^{2\alpha_1} \exp[(1 - \alpha_1)\alpha_0] \exp[g(\epsilon_{h+1})].$$

Taking conditional expectation at time h , we have

$$\hat{\sigma}_h^2(2) = \hat{\sigma}_h^{2\alpha_1}(1) \exp[(1 - \alpha_1)\alpha_0] E_h\{\exp[g(\epsilon_{h+1})]\},$$

where E_h denotes a conditional expectation taken at the time origin h . The prior expectation can be obtained as follows:

$$\begin{aligned} E\{\exp[g(\epsilon)]\} &= \int_{-\infty}^{\infty} \exp[\theta\epsilon + \gamma(|\epsilon| - \sqrt{2/\pi})] f(\epsilon) d\epsilon \\ &= \exp\left(-\gamma\sqrt{2/\pi}\right) \left[\int_0^{\infty} e^{(\theta+\gamma)\epsilon} \frac{1}{\sqrt{2\pi}} e^{-\epsilon^2/2} d\epsilon \right. \\ &\quad \left. + \int_{-\infty}^0 e^{(\theta-\gamma)\epsilon} \frac{1}{\sqrt{2\pi}} e^{-\epsilon^2/2} d\epsilon \right] \\ &= \exp\left(-\gamma\sqrt{2/\pi}\right) \left[e^{(\theta+\gamma)^2/2} \Phi(\theta + \gamma) + e^{(\theta-\gamma)^2/2} \Phi(\gamma - \theta) \right], \end{aligned}$$

where $f(\epsilon)$ and $\Phi(x)$ are the probability density function and CDF of the standard normal distribution, respectively. Consequently, the 2-step ahead volatility forecast is

$$\begin{aligned} \hat{\sigma}_h^2(2) &= \hat{\sigma}_h^{2\alpha_1}(1) \exp\left[(1 - \alpha_1)\alpha_0 - \gamma\sqrt{2/\pi}\right] \\ &\quad \times \{\exp[(\theta + \gamma)^2/2] \Phi(\theta + \gamma) + \exp[(\theta - \gamma)^2/2] \Phi(\gamma - \theta)\}. \end{aligned}$$

Repeating the previous procedure, we obtain a recursive formula for a j -step ahead forecast:

$$\begin{aligned} \hat{\sigma}_h^2(j) &= \hat{\sigma}_h^{2\alpha_1}(j-1) \exp(\omega) \\ &\quad \times \{\exp[(\theta + \gamma)^2/2] \Phi(\theta + \gamma) + \exp[(\theta - \gamma)^2/2] \Phi(\gamma - \theta)\}, \end{aligned}$$

where $\omega = (1 - \alpha_1)\alpha_0 - \gamma\sqrt{2/\pi}$. The values of $\Phi(\theta + \gamma)$ and $\Phi(\theta - \gamma)$ can be obtained from most statistical packages. Alternatively, accurate approximations to these values can be obtained by using the method in Appendix B of Chapter 6.

For illustration, consider the AR(1)–EGARCH(1,1) model of the previous subsection for the monthly log returns of IBM stock, ending December 1997. Using the fitted EGARCH(1,1) model, we can compute the volatility forecasts for the series. At the forecast origin $t = 864$, the forecasts are $\hat{\sigma}_{864}^2(1) = 6.05 \times 10^{-3}$, $\hat{\sigma}_{864}^2(2) = 5.82 \times 10^{-3}$, $\hat{\sigma}_{864}^2(3) = 5.63 \times 10^{-3}$, and $\hat{\sigma}_{864}^2(10) = 4.94 \times 10^{-3}$. These forecasts converge gradually to the sample variance 4.37×10^{-3} of the shock process a_t of Eq. (3.29).

3.9 THE THRESHOLD GARCH MODEL

Another volatility model commonly used to handle leverage effects is the threshold GARCH (or TGARCH) model; see Glosten, Jagannathan, and Runkle (1993) and Zakoian (1994). A TGARCH(m, s) model assumes the form

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^s (\alpha_i + \gamma_i N_{t-i}) a_{t-i}^2 + \sum_{j=1}^m \beta_j \sigma_{t-j}^2, \quad (3.33)$$

where N_{t-i} is an indicator for *negative* a_{t-i} , that is,

$$N_{t-i} = \begin{cases} 1 & \text{if } a_{t-i} < 0, \\ 0 & \text{if } a_{t-i} \geq 0, \end{cases}$$

and α_i , γ_i , and β_j are non-negative parameters satisfying conditions similar to those of GARCH models. From the model, it is seen that a positive a_{t-i} contributes $\alpha_i a_{t-i}^2$ to σ_t^2 , whereas a negative a_{t-i} has a larger impact $(\alpha_i + \gamma_i) a_{t-i}^2$ with $\gamma_i > 0$. The model uses zero as its *threshold* to separate the impacts of past shocks. Other threshold values can also be used; see Chapter 4 for the general concept of threshold models. Model (3.33) is also called the GJR model because Glosten et al. (1993) proposed essentially the same model.

For illustration, consider the monthly log returns of IBM stock from 1926 to 2003. The fitted TGARCH(1,1) model with conditional GED innovations is

$$\begin{aligned} r_t &= 0.0121 + a_t, \quad a_t = \sigma_t \epsilon_t, \\ \sigma_t^2 &= 3.45 \times 10^{-4} + (0.0658 + 0.0843 N_{t-1}) a_{t-1}^2 + 0.8182 \sigma_{t-1}^2, \end{aligned} \quad (3.34)$$

where the estimated parameter of the GED is 1.51 with standard error 0.099. The standard error of the parameter for the mean equation is 0.002 and the standard errors of the parameters in the volatility equation are 1.26×10^{-4} , 0.0314, 0.0395, and 0.049, respectively. To check the fitted model, we have $Q(12) = 18.34(0.106)$ for the standardized residual $\tilde{a}_t$ and $Q(12) = 5.36(0.95)$ for $\tilde{a}_t^2$. The model is adequate in modeling the first two conditional moments of the log return series. Based on the fitted model, the leverage effect is significant at the 5% level.

S-Plus Commands Used

```
> ibm.tgarch = garch(ibm.ln~1, ~tgarch(1,1), leverage=T,
+ cond.dist='ged')
> summary(ibm.tgarch)
> plot(ibm.tgarch)
```